An Introduction to OEIS Sequence A395620

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Abstract

This paper presents an introduction to the integer sequence A395620 from the Online Encyclopedia of Integer Sequences (OEIS). The sequence counts permutations of the set $1, 2, \dots, n$ whose prefix sums never form a perfect square. This problem lies at the intersection of combinatorics and number theory and provides an interesting example of how arithmetic constraints affect the structure of permutations.

1 Introduction

Integer sequences play a fundamental role in mathematics, providing insight into combinatorial structures, number theoretic phenomena, and algorithmic processes. One particularly interesting sequence is OEIS A395620, which counts permutations that avoid perfect squares among their partial sums.

The study of constrained permutations has become an important area of combinatorics. By imposing arithmetic restrictions on permutations, mathematicians can investigate how local numerical conditions influence global combinatorial behavior.

2 Definition of the Sequence

Consider the set

1 , 2 , łd o t s , n .

A permutation of this set can be written as

(p 1 , p 2 , łd o t s , p n) .

For each permutation, define the prefix sums

S k = p 1 + p 2 + d ¸ o t s + p k ,

where

1 łe q k łe q n .

The permutation is considered valid if none of the values S_k is a perfect square.

Therefore, A395620 counts all permutations satisfying the condition

S k e q m 2

for every positive integer m and every index k.

3 Examples

To illustrate the concept, consider the case $n = 2$.

The possible permutations are

(1 , 2)

and

(2 , 1) .

For the permutation $(1, 2)$, the prefix sums are

1 , u a d 3.

Since $1 = 1^2$ is a perfect square, this permutation is rejected.

For the permutation $(2, 1)$, the prefix sums are

2 , u a d 3.

Neither value is a perfect square, so this permutation is accepted.

Thus, only one valid permutation exists in this case.

For larger values of n, the number of valid permutations grows rapidly, although many arrangements must still be discarded because they generate square prefix sums.

4 Mathematical Significance

Sequence A395620 is interesting because it combines two classical mathematical themes.

The first is combinatorics, since the problem involves counting permutations. The second is number theory, since the restriction is based on perfect squares.

The challenge arises from the fact that the condition must hold throughout the entire permutation. A single prefix sum equal to a square is enough to invalidate the arrangement.

This type of problem belongs to a broader class of avoidance problems, where mathematical objects are counted only if they avoid certain forbidden patterns or numerical properties.

5 Growth of the Sequence

The first terms of the sequence are

1 , 0 , 1 , 3 , 6 , 46 , 220 , 1540 , łd o t s

As n increases, the total number of permutations grows as

n ! ,

which increases extremely quickly.

Although many permutations are excluded by the square-avoidance condition, a substantial number remain valid. Understanding the asymptotic behavior of the sequence remains an interesting combinatorial question.

6 Conclusion

OEIS sequence A395620 provides an elegant example of the interaction between combinatorics and number theory. By counting permutations whose prefix sums avoid perfect squares, the sequence illustrates how arithmetic constraints can significantly influence combinatorial structures.

Future investigations may focus on efficient counting algorithms, asymptotic estimates, and connections with other classes of restricted permutations.

References

[1] The OEIS Foundation Inc., *Sequence A395620*, Online Encyclopedia of Integer Sequences.

[2] Richard P. Stanley, *Enumerative Combinatorics*, Cambridge University Press.

[3] Martin Aigner, *A Course in Enumeration*, Springer.